Delft University of Technology

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Aircraft Tutorial

Group 30
Ravindu Kalinga Bandara (5958016)
Alexandru-Theodor Minculescu (5983681)
Cristian Caba (6022537)
Selim Kazanci (5912261)
Daniel Constantinescu (5908345)
Alin Nagy (6004598)

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1 Introduction

The report investigates the effects of propulsive solutions on the characteristics of a regional aircraft, using the Bombardier CRJ-1000 as the reference configuration. The study is carried out in the context of a conceptiual redesign in which the aircraft is transformed into a hybrid-electric variant, reffert to as CRJ-EXX. The objective of the report is to assess how the proposed modifications influence the aircraft from a weight and balance, stability point of view.

The first part of the report focuses on establishing the main characteristics of the CRJ-1000. This includes the determination of the aircraft weight breakdown, the estimation of the OEW center of gravity, the construction of the loading diagram and the

The second part examines the CRJ-EXX configuration, which includes several important modifications, including a hybrid-electric propulsion system, battery packs placed in the cargo holds, structural mass reductions, improved high lift devices and increased the effective aspect ratio. (reduced capacity in passengers). These changes affect both the mass distribution and the aerodynamic characteristics of the aircraft, therefore ... The modified aircraft is examined using the same framework as the baseline aircraft, allowing direct comparison between the two cases.

2 Analysis of CRJ-1000

2.1 General Information about CRJ-1000

show the calculation for cg of cargo holds

In order to assess the stability and controllability (S&C) of CRJ-1000, certain parameters were required for analysis where they were used at different places throughout this report. In this subsection, the **relevant** parameters are introduced, if calculations were done to obtain certain parameters they were presented and the assumptions made are explained. The parameters that do not concern the S&C such as engine thrust capabilities etc. were not shown due to irrelevancy.
The Table 1 displays the general parameters of the aircraft whereas the Table 2 shows the aerodynamic parameters.

Table 1: Aircraft Input Parameters: Performance, Payload, Fuselage, and Landing Gear[1] [2]

Parameter	Value	Unit	Parameter	Value	Unit
Payload & Cabin			**Fuselage & Cockpit Dimensions**		
Passenger Mass (incl. bag)	87	kg	Fuselage Length	36.47	m
Front Cargo Mass	472	kg	Fuselage Max Diameter	4.0	m
Front Cargo CG Position	8.3	m	Fuselage CG Ratio	0.45	–
Aft Cargo Mass	1,294	kg	Cockpit x-Position	2.5	m
Aft Cargo CG Position	26.5	m	Reference z-Position	4.0	m
Number of Seat Rows	25	–	**Landing Gear Options** (*Wheels*)		
Row 1 Position	7.1	m	Nose Gear Station	2.126	m
Seat Pitch	0.79	m	Main Gear Station	22.10	m
Flight Characteristics & Limits			Main Gear Span (Half-Track)	3.0	m
Cruise Mach Number	0.82	–	Number of Main Gears	2	–
Cruise Altitude	9,000	m	Tail Strike Angle	15	deg

Table 2: Aircraft Input Parameters: Aerodynamics, Surfaces, and Engine Nacelles[1] [3]

Parameter	Value	Unit	Parameter	Value	Unit
Main Wing Dimensions			**Horizontal Stabiliser Dimensions**		
Root Chord	5.12	m	Root Chord	2.58	m
Wingspan	26.2	m	Tip Chord	1.10	m
Wing Area	77.4	m^2	Planform Area	15.91	m^2
Taper Ratio	0.3	–	Span	8.65	m
Leading Edge Sweep Angle	29.5	deg	Taper Ratio	0.426	–
LEMAC Position	19.75	m	Sweep Angle	33.7	deg
Front Spar Fraction	0.20	–	LEMAC Position	36.7	m
Rear Spar Fraction	0.80	–			
Flap Start Position	1.289	m			
Flap End Position	9.79	m			
Engine & Nacelles			**Vertical Stabiliser Dimensions**		
Nacelle Start Position	30.0	m	Root Chord	3.27	m
Nacelle Lateral Offset	3.0	m	Tip Chord	2.21	m
Nacelle Length	3.9	m	Planform Area	11.32	m^2
Nacelle Max Diameter	1.66	m	Span	4.0	m
Number of Nacelles	2	–	Taper Ratio	0.676	–
			Sweep Angle	42.6	deg
Aerodynamic Coefficients			LEMAC Position	35.0	m
Max Lift Coefficient	2.6	–	Root z-Position	1.35	m
Zero-Lift Pitching Moment	-0.08	–			
Zero-Lift Lift Coefficient	0.05	–			
c_f	0.7	–			
c_s	0.58	–			

The passenger mass[4] was taken from the upper end for sake of conservative assumption, in case of a male-passenger dominated flight.
The locations of the cargo holder cg were calculated as the midpoints of the cargo holder [1], therefore, it was assumed a uniform cargo distribution throughout the cargo hold volume.
For the seat pitch, it was assumed that the seat and passenger distribution is uniform with no differences, hence, no emergency exit seat pitch etc. were taken into account. All the seats were placed right after each other.

2.1.1 Aerodynamic surface size approximation

Since the vast majority of the aerodynamic surface sizing are not publicly available, and instead only present in specific, classified Aircraft Maintenance Manuals, the size of the aerodynamic surfaces and positions and sizes of elements, such as nacelles and landing gear. Two accurate, to-scale CAD drawings of the CRJ-1000 were used in this process, as it accurately represents the real dimensions of the airplane (). The first step was measuring the fuselage, which had a length of 566 pixels in Figure 1 and a length of 446 pixels in Figure 2, while its real documented length is $36.47m$. Thus, it was determined that Figure 1 had a scale of $0.064435m$ per pixel, while Figure 2 had a scale of $0.08177m$ per pixel. Angles were also measured using digital protractors. All element positions have been measured from the tip of the nose of the aircraft except for the flap start and end positions, which were measured from the center of fuselage. In Table 3, all sizes estimated using this method can be found, together with their pixel and estimated real size.

cite the cad drawing

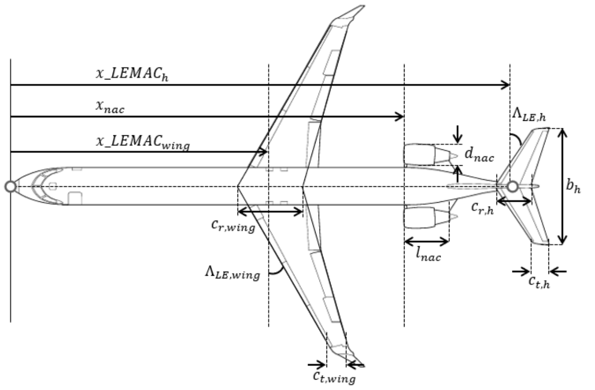

Figure 1: Top-down CAD drawing of CRJ-1000

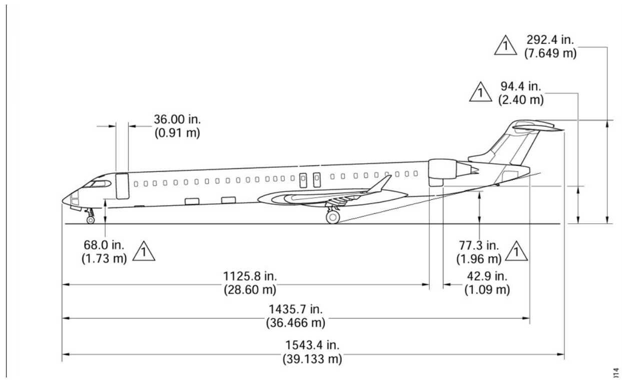

Figure 2: Side-view CAD drawing of CRJ-1000

Table 3: Table of estimated sizes, lengths and angles

Figure 1			Figure 2		
Element	# of pixels	Size	Element	# of pixels	Size
Horizontal Tail MAC x-location	582	37.5m	Nose wheel x-location	26	2.13m
Horizontal Tail Root chord	37	2.36m	Main wheel x-location	270	22.1m
Horizontal Tail Tip chord	17	1.1m	Vertical stabilizer root chord	40	3.27m
Horizontal Tail Span	143	9.2m	Vertical stabilizer tip chord	27	2.21m
Wing root chord	80	5.12	Tail-Strike Angle	N.A.	15 °
Nacelle start x-location	466	30m			
Nacelle Length	60	3.9m			
Nacelle Width	26	1.66m			
Horizontal tail sweep	N.A.	33.7°			
Wing sweep	N.A.	29.5 °			

It is important to note that while wing root chord and wing tip chord were not values publicly available online, the wing surface area was. However, while considering the wing a trapezoid, the computed wing area from root and tip chord and the publicly available wing area value do not match. This is because, in reality, as can be seen in Figure 1, Bombardier aircraft designers decided to add an extra spar near the root of the wing to accommodate the main landing gear. For this reason, the actual effective wing surface has a higher value.
The taper ratios were calculated by $\frac{c_r}{c_t}$ where c_r is the chord length at the root and c_t is the chord length at the tip of the aerodynamic surface.
Maximum lift coefficient of the aircraft was calculated from the Figure 3 as shown below

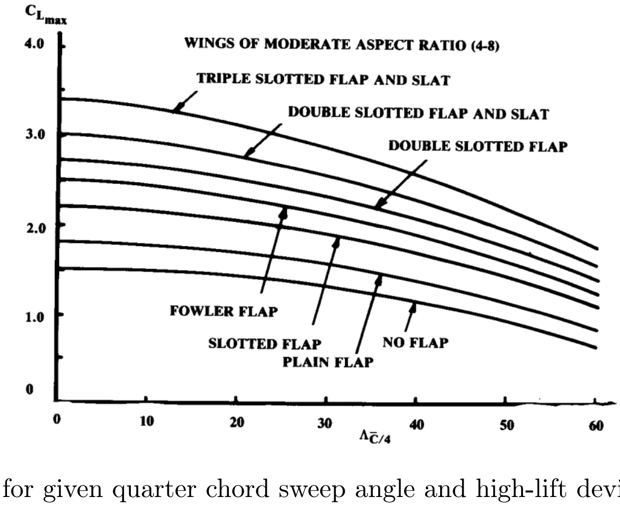

Figure 3: Maximum C_L for given quarter chord sweep angle and high-lift device configuration of the wing

Where the quarter chord sweep angle was calculated by $\Lambda_{c/4} = \arctan\left(\tan(\Lambda_{LE}) - \frac{1-\lambda}{AR(1+\lambda)}\right)$ which gave the value of 26.9 deg and the type of high lift devices CRJ-1000 uses is categorized as double slotted flap and slat. Thus, for these inputs, reading from the graph, maximum C_L of 2.6 was obtained.

Zero-lift pitching moment coefficient of the airfoil was found from a generic analysis of supercritical airfoils[5], which CRJ-1000s airfoils also fall under this category.

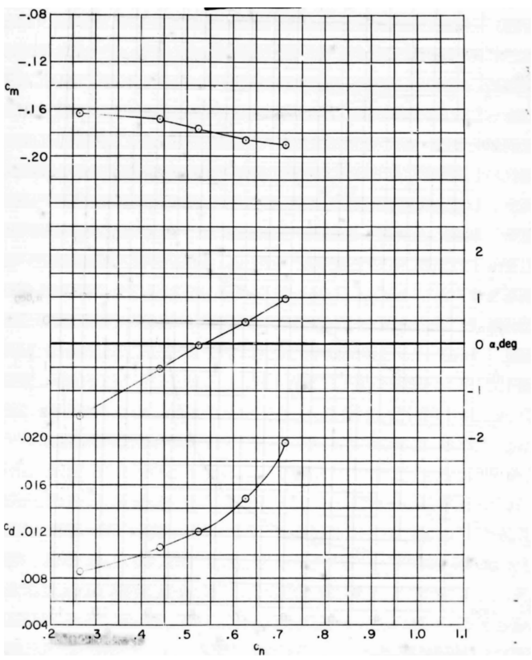

Figure 4: C_N against C_m curve for supercritical airfoils at $M = 0.77$

It was assumed that $C_N \approx C_L$ and $M = 0.77$ is close to the cruise Mach number of CRJ-1000 thus, reading from the graph, C_m value at zero C_N is found to be -1.6 which corresponds to the zero lift moment coefficient.

The zero lift coefficient C_{L_0} was found by empirical methods[6]. The following parameters were used for the aircraft operating at a cruise Mach of 0.82 where the remaining new values were assumed to be the standard values for this analysis for aircraft similar to CRJ-1000:

- Aspect Ratio (A): 8.85
- Half-chord Sweep ($\Lambda_{1/2}$): 22°
- Taper Ratio (λ): 0.3
- Cruise Mach Number (M): 0.82
- Wing Incidence (i_w): 0.0°
- Tip Twist/Washout (ϵ_t): $-1.5°$
- Airfoil Zero-Lift Angle (α_{0l}): $-1.2°$
- Airfoil Efficiency Factor (κ): 0.95
- Tail Trim Downforce ($C_{L_{tail}}$): -0.15

To account for transonic cruise speeds, the compressibility factor (β) is calculated:

$$\beta = \sqrt{1 - M^2} \tag{1}$$

The three-dimensional lift curve slope of the wing is calculated using the modified Helmbold equation for swept wings in compressible flow:

$$C_{L_\alpha} = \frac{2\pi A}{2 + \sqrt{4 + \frac{A^2\beta^2}{\kappa^2}\left(1 + \frac{\tan^2 \Lambda_{1/2}}{\beta^2}\right)}} \tag{2}$$

The effective zero-lift angle of the 3D wing accounts for the aerodynamic washout (twist) distributed across the span. For a straight-tapered wing, this is integrated as:

$$\alpha_{0L_w} = \alpha_{0l} + \epsilon_t \left(\frac{1+2\lambda}{3+3\lambda} \right) \tag{3}$$

The lift generated by the wing when the fuselage is at zero degrees angle of attack is a function of its mounting incidence and its zero-lift angle:

$$C_{L_{0_{wing}}} = C_{L_\alpha}(i_w - \alpha_{0L_w}) \tag{4}$$

To maintain level flight, the horizontal tail must generate a downforce to counteract the pitching moment and balance the forward center of gravity. The total aircraft C_{L_0} is the sum of the wing's lift and the tail's trim penalty:

$$C_{L_{0_{total}}} = C_{L_{0_{wing}}} + C_{L_{tail}} \tag{5}$$

Plugging in all the parameters results in

$$C_{L_{0_{total}}} = 0.199 + (-0.15) \approx 0.05 \tag{6}$$

Thus, at a cruise Mach of 0.82 with the specified trim and incidence parameters, the estimated zero-lift lift coefficient for the aircraft is approximately 0.05.

write the calculation of c and c_s

2.2 Empty operative weight breakdown and CG location

Before constructing the loading diagram, the center of gravity (CG) of the aircraft in the Empty Operating Weight (EOW) configuration must be determined. The weight distribution of the components contributing to the EOW was established in Task 1 and is summarized in Table 4.

Table 4: Weight Breakdown and CG Locations

Component	**Mass (% EOW)**	**Mass (kg)**	**CG from Nose (m)**	**CG from LEMAC (m)**	**CG as % MAC**
Wing	19.7	4568.04	21.70	1.95	53.43
Horizontal Tail	2.5	579.70	37.51	17.76	486.75
Vertical Tail	1.8	417.38	36.17	16.42	449.77
Fuselage (incl. cabin and furnishing systems)	35.0	8115.80	16.41	-3.34	-91.47
Main Landing gear	5.8	1344.90	22.10	2.35	0.64
Nose Landing gear	0.8	185.50	2.13	-17.62	-482.79
Propulsion System (including Nacelles)	13.3	3084.00	31.56	11.81	323.59
Cockpit Systems (avionics, furnishing...)	2.3	533.32	2.5	-17.25	-472.65
Total Accounted EOW	**81.2**	**18828.66**	**21.13**	**1.38**	**37.95**

The remaining 18.8% of the EOW, not included in Table 4, was assumed to have no influence on the CG, as specified in the task. The CG of a group of components, whether the fuselage group, wing group, or the entire aircraft, was calculated using:

$$x_{cg} = \frac{\sum x_i m_i}{\sum m_i} \tag{7}$$

The fuselage group comprises the horizontal and vertical tails, fuselage (including cabin and furnishing systems), nose landing gear, propulsion system (including nacelles), and cockpit systems. The propulsion system is included in this group because the CRJ-1000 features fuselage-mounted engines.

The wing group consists of the wing and the main landing gear. The main landing gear is included in this group as it influences both the CG location and the allowable CG limits due to ground clearance and stability considerations during ground operations. Referring the component masses and CG locations in Table 4, the CG locations of the fuselage group, wing group and finally the whole aircraft in EOW configuration can be calculated as show in in Table 5.

Table 5: Assembly Group Weight Breakdown and CG Locations

Assembly Group	Mass (kg)	CG from Nose (m)	CG from LEMAC (m)	CG as % MAC
Fuselage Group	12915.72	20.83	1.08	29.72
Wing Group	5912.94	21.79	2.04	55.92
Complete Aircraft at EOW	**23188**	**21.13**	**1.38**	**37.95**

2.3 Weight chart for CRJ-1000

This subsection introduces the CRJ-1000 weight data and how the weight of the aircraft is split.

Table 6: CRJ-1000 Weight Breakdown [3]

Weight Component	Mass [kg]	% of MTOW
Maximum Takeoff Weight (MTOW)	**41,640**	**100.0%**
Operational Empty Weight (OEW)	23,188	55.7%
Fuel Weight (@ Max Payload)	6,495	15.6%
Maximum Payload Weight	11,957	28.7%
Payload Weight Split:		
1) Pax & Cabin Luggage	8,700	20.9%
2) Front Cargo Hold	870	2.3%
3) Aft Cargo Hold	2,384	5.5%

The maximum cargo hold weight[3] was 3257 kg which then split into $\frac{5.26}{19.67} = 26.7\%$ and $\frac{14.41}{19.67} = 73.2\%$ between the forward and aft cargo holds respectively, according to their internal volume capacity [1] where it was assumed that the cargo density was same for both cargo holds.
Thus, maximum zero fuel weight (MZFW) was found to be $MZFW = OEW + MPW \Rightarrow 23188 + 11957 = 35145$kg.
Finally, the maximum fuel weight at maximum payload was calculated by $MTOW - MZFW \Rightarrow 41640 - 35145 = 6495$kg.
Gathering all these, the following pie chart was created as shown below in Figure 5

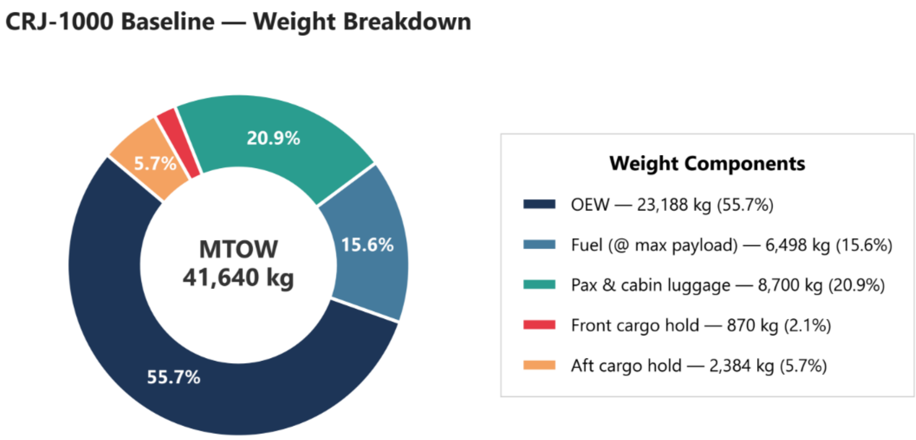

Figure 5: Pie chart of the weight distribution of CRJ-1000

2.4 Loading Diagram of CRJ-1000

In this subsection, the generation of the aircraft loading diagram, which is also referred to as potato diagram is presented. Potato diagram was a critical tool for determining the operational limits of CRJ-1000 . The physical shift in the aircraft's CG as varying combinations of payload and fuel were added. By plotting aircraft mass against the CG position (expressed as a percentage of the MAC), the absolute forward and aft physical limits of the aircraft were established, which were then used in . refer to the scissor plot chapter

As discrete masses (passengers, cargo, or fuel) were added incrementally where the required weight and CG location parameters were calculated in subsection 2.1, subsection 2.3 and **??**. The new aircraft mass W_{new} and new CG location $x_{cg,new}$ were calculated using the standard weighted-average formula for each iteration:

$$W_{new} = W_{old} + W_{added} \tag{8}$$

$$x_{cg,new} = \frac{(W_{old} \cdot x_{cg,old}) + (W_{added} \cdot x_{cg,added})}{W_{new}} \tag{9}$$

To determine the absolute boundaries of the envelope, two parallel loading paths were simulated:

- **Path A (Forward-to-Aft Loading):** This path simulates the scenario where the aircraft is loaded strictly from the front to the back, driving the CG as far forward as possible at intermediate weights. The sequence is:
  1. Front cargo hold loaded to maximum capacity.
  2. Aft cargo hold loaded to maximum capacity.
  3. Window seats loaded progressively from Row 1 to the final row.
  4. Aisle seats loaded progressively from Row 1 to the final row.
- **Path B (Aft-to-Forward Loading):** This path reverses the loading order to drive the CG as far aft as possible. The sequence is:
  1. Aft cargo hold loaded to maximum capacity.
  2. Front cargo hold loaded to maximum capacity.
  3. Window seats loaded progressively from the final row up to Row 1.
  4. Aisle seats loaded progressively from the final row up to Row 1.

The cargo were assumed to be single uniform boxes and as it was explained, their CG location was assumed to be at the midpoint of the cargo hold due to uniform cargo mass distribution. Furthermore, CG location of the passengers was assumed to be the position of the seat. Once both payload paths were fully loaded, the aircraft reached its MZFW configuration.
Finally, the fuel was added, where the CG location of the fuel was assumed to be the geometrical CG location of the wing. This assumption is fairly valid since the fuel was assumed to be stored only at the wings and it was distributed homogeneous along the wing volume. Thus, this location corresponded to the geometrical CG location of the wing, which was found in **??**. The resulting diagram is shown below in Figure 6.

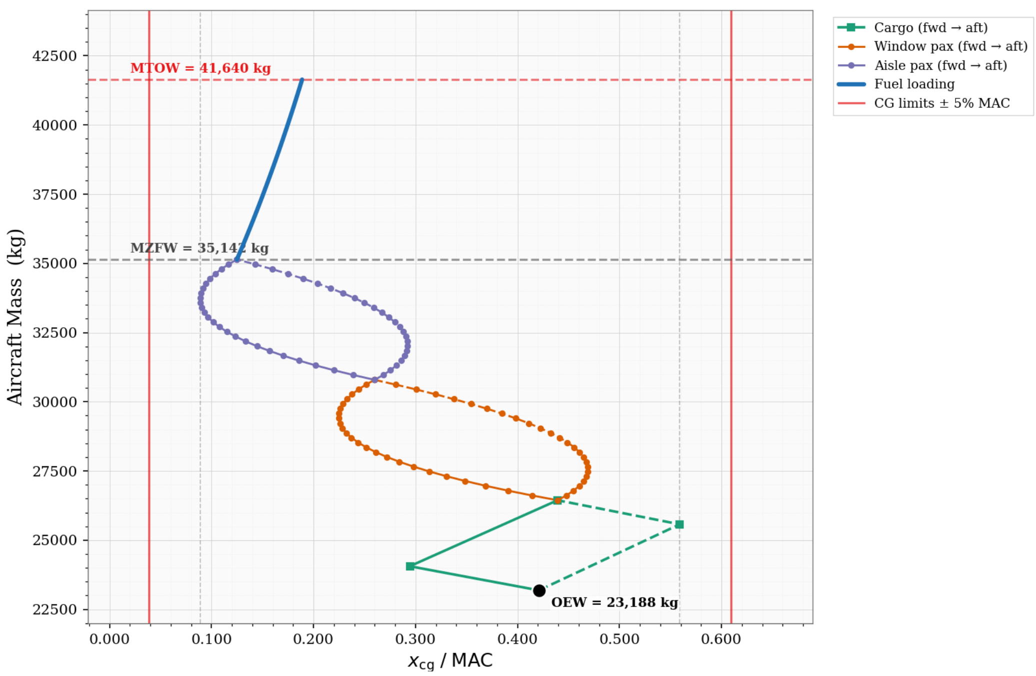

Figure 6: CRJ-1000 Loading Diagram

Lastly, from these combinations, the algorithm extracts the most forward and most aft CG coordinates generated at any point during the loading process. Finally, a safety margin of $\pm5\%$ MAC is applied to both extremes. This operational margin is vital, as it accounts for passenger movement, loading inaccuracies, and fuel slosh, ensuring the aircraft remains controllable under all circumstances.
The most forward calculated CG occurred at exactly $\approx 8.87\%$ MAC, driven by the forward-loading sequence (Path A) before the influence of aft payloads. Conversely, the most aft CG was identified at $\approx 55.91\%$ MAC, generated during the aft-loading sequence (Path B). Moreover, after applying the safety margin, the final permitted Operational CG Envelope for the CRJ-1000 is defined between a forward limit of **3.87% MAC** and an aft limit of **60.91% MAC**.
When the shape of the loading diagram is investigated, the result is as expected. Due to aft wing and engine configurations, the OEW CG location was also aft. Furthermore, the heavier, aft cargo hold caused the extreme aft CG limit as expected, since the majority of the passengers were located in front of the aircraft. Thus, the extreme forward CG location was due to the passengers since as discussed, the fuel was stored at the wings which were configured aft of the aircraft.

2.5 Stability and Control

2.5.1 Stability

To assess an aircraft's longitudinal characteristics, a scissor plot is constructed to evaluate both static stability and controllability. A scissor plot is a graphical tool that combines the stability and controllability constraints by plotting both requirements as functions of the CG location. In practice, these constraints are expressed as limiting curves and plotted against the tail-to-wing area ratio. For each value of this ratio, the corresponding forward and aft CG limits are obtained. This representation illustrates how the allowable CG range varies with tail size and enables the selection of a configuration that satisfies both stability and controllability requirements across the desired CG range.

Static longitudinal stability describes how the aircraft responds to a small perturbation in angle of attack. An aircraft is statically stable if it generates a restoring pitching moment that returns it to equilibrium. If a disturbance instead causes the pitching moment to diverge further from equilibrium, the aircraft is statically unstable. Generally, the stability line of the scissor plot defines the aft limit of the CG position, given by:

$$\bar{x}_{np} = \bar{x}_{ac} + \frac{C_{L_{\alpha_h}}}{C_{L_{\alpha_{A-h}}}}\left(1-\frac{d\epsilon}{d\alpha}\right)\frac{S_h l_h}{S\bar{c}}\left(\frac{V_h}{V}\right)^2 \quad\Rightarrow\quad \bar{x}_{cg} = \bar{x}_{ac} + \frac{C_{L_{\alpha_h}}}{C_{L_{\alpha_{A-h}}}}\left(1-\frac{d\epsilon}{d\alpha}\right)\frac{S_h l_h}{S\bar{c}}\left(\frac{V_h}{V}\right)^2 - S.M. \tag{10}$$

The overbar notation (e.g., $\bar{x}$) indicates that positions are expressed relative to the leading edge of the mean aerodynamic chord (MAC) and normalized by the MAC length. In Equation 10, $\bar{x}_{np}$ denotes the non-dimensional neutral point location, $\bar{x}_{cg}$ the non-dimensional CG location, and $\bar{x}_{ac}$ the non-dimensional aerodynamic center location of the aircraft. The terms $C_{L_{\alpha_h}}$ and $C_{L_{\alpha_{A-h}}}$ represent the lift curve slopes of the horizontal tail and the aircraft minus the horizontal plane, respectively. The downwash gradient, $\frac{d\epsilon}{d\alpha}$, accounts for the reduction in effective angle of attack at the tail due to the wing. l_h is the tail moment arm (distance between the wing and tail aerodynamic centers). The term $\left(\frac{V_h}{V}\right)^2$ represents the ratio of the airflow velocity at the tail relative to that at the wing.

The static margin, denoted by $S.M.$, represents the distance between the CG and the neutral point and serves as a safety margin to ensure sufficient longitudinal stability. It is introduced not only as a design buffer but also to account for the difference between stick-fixed and stick-free stability conditions. While the neutral point derived here corresponds to the stick-fixed case, the stick-free condition is typically more restrictive, often shifting the neutral point forward. Therefore, a positive static margin is required to guarantee stable behavior under all conditions.

In order to utilize Equation 10, it is necessary to estimate the parameters $C_{L_{\alpha_h}}$, $C_{L_{A-h}}$, $\left(\frac{V_h}{V}\right)$, and $\frac{d\epsilon}{d\alpha}$ for the most critical flight condition. For longitudinal stability considerations, this condition is typically the cruise flight state, where the aircraft operates for the majority of its mission and stability requirements are most representative.

1. $\left(\mathbf{\frac{V_h}{V}}\right)$

   Since the aircraft has a T-tail configuration, there is no loss of velocity at the horizontal stabaliser due to the wing, $\frac{V_h}{V} = 1$.

2. $\mathbf{C_{L_{\alpha_h}}}$

Parameter	Variable	Value
Tail Aspect Ratio	A_h	5.316
Tail Half-Chord Sweep	$\Lambda_{h,c/2}$	27.92°
Tail Velocity Ratio	V_h/V	1.0
Tail Mach Number	M_h	0.82
Tail Compressibility Factor	β_h	0.5724

Table 7: Fundamental parameters used for the Tail Lift Curve Slope ($C_{L_{\alpha_h}}$) calculation.

$C_{L_{\alpha_h}}$ was calculated using Equation 2, but using the horizontal stabilizer parameters listed in Table 7. The Mach correction factor is identical to that of the wing since for a T-tail configuration $\frac{V_h}{V} = 1$. This results in $C_{L_{\alpha_h}} = 4.9111\,\text{rad}^{-1}$.

3. $\mathbf{C_{L_{\alpha_{A-h}}}}$

Parameter	Variable	Value
Fuselage Width / Diameter	b_f (or d_{fus})	4 m
Wing Span	b_w	26.2 m
Wing Area	S_w	77.4 m^2
Wing Root Chord	c_{rw}	5.12 m
Wing Aspect Ratio	A_w	8.85
Wing Half-Chord Sweep	$\Lambda_{w,c/2}$	22°
Cruise Mach Number	M	0.82
Wing Compressibility Factor	β_w	0.5724
Airfoil Efficiency Factor	κ	0.95
Fuselage Interference Constant	–	2.15

Table 8: Fundamental parameters and empirical constants used for the Aircraft-less-tail lift curve slope ($C_{L_{\alpha_{A-h}}}$) calculation.

The lift curve slope of the aircraft without the horizontal tail, $C_{L_{\alpha_{A-h}}}$, is obtained by correcting the isolated wing lift curve slope to account for fuselage interference and the reduction in effective lifting area due to the wing portion enclosed within the fuselage.

The lift curve slope of the isolated wing, $C_{L_{\alpha_w}}$, is calculated using the DATCOM method (Equation 2), which yields an intermediate value of $C_{L_{\alpha_w}} = 6.1546$ rad^{-1}.

First, the fuselage interference factor is introduced to account for the aerodynamic interaction between the fuselage and the wing root:

$$K_{int} = 1 + 2.15 \left(\frac{b_f}{b_w} \right) \tag{11}$$

where b_f is the fuselage width and b_w is the wing span.

Next, the net wing area exposed to the freestream is computed by subtracting the portion of the wing enclosed within the fuselage:

$$S_{net} = S_w - b_f c_{rw} \tag{12}$$

where S_w is the reference wing area.

The lift curve slope of the wing-body combination is then expressed as:

$$C_{L_{\alpha_{A-h}}} = C_{L_{\alpha_w}} K_{int} \left(\frac{S_{net}}{S_w} \right) + \frac{\pi}{2} \left(\frac{b_f^2}{S_w} \right) \tag{13}$$

The first term represents the corrected wing contribution, while the second term accounts for the additional lift generated by the fuselage itself.

Substituting the intermediate wing value and the parameters from Table 8 yields $C_{L_{\alpha_{A-h}}} \approx 6.5795$ rad^{-1}.

4. $\mathbf{\frac{d\epsilon}{d\alpha}}$

Parameter	Variable	Value
Wing Leading Edge Sweep	Λ_{LE_w}	29.5°
Wing Taper Ratio	λ_w	0.3
Tail Horizontal Position Parameter	r	1.335 m
Tail Vertical Position Parameter	m_{tv}	0.359 m

Table 9: Fundamental parameters used for the downwash derivative ($\frac{\partial\epsilon}{\partial\alpha}$) calculation.

The downwash derivative, $\frac{\partial\epsilon}{\partial\alpha}$, quantifies the rate of change of the airflow deflection angle at the horizontal tail caused by the wing's wake. This is calculated using empirical DATCOM formulations that account for the relative distance between the wing and the tail, as well as the wing's sweep and aspect ratio.

The calculation relies on the isolated wing lift curve slope, which was previously determined to be $C_{L_{\alpha_w}} = 6.1546\ \text{rad}^{-1}$, as well as the wing aspect ratio (A_w) defined in Table 8.

First, the sweep correction factors ($K_{\epsilon,\Lambda}$ and K_{ϵ,Λ_0}) are computed based on the wing's quarter-chord sweep angle ($\Lambda_{w,c/4}$) and the normalized horizontal distance to the tail (r):

$$K_{\epsilon,\Lambda} = \frac{0.1124 + 0.1265\Lambda_{w,c/4} + 0.1766\Lambda^2_{w,c/4}}{r^2} + \frac{0.1024}{r} + 2 \tag{14}$$

$$K_{\epsilon,\Lambda_0} = \frac{0.1124}{r^2} + \frac{0.1024}{r} + 2 \tag{15}$$

The downwash derivative is then calculated by combining the sweep corrections with the vertical and horizontal tail positioning parameters (m_{tv} and r), scaled by the wing's lift curve slope over its aspect ratio:

$$\frac{\partial\epsilon}{\partial\alpha} = \left(\frac{K_{\epsilon,\Lambda}}{K_{\epsilon,\Lambda_0}}\right)\left[\frac{r}{r^2+m_{tv}^2}\frac{0.4876}{\sqrt{r^2+0.6319+m_{tv}^2}} + \left(1+\left(\frac{r^2}{r^2+0.7915+5.0734m_{tv}^2}\right)^{0.3113}\right)\left(1-\sqrt{\frac{m_{tv}^2}{1+m_{tv}^2}}\right)\right]\frac{C_{L_{\alpha_w}}}{\pi A_w} \tag{16}$$

Substituting the geometric parameters from Table 9 and Table 8 yields the final downwash derivative $\frac{\partial\epsilon}{\partial\alpha} \approx 0.3232$.

5. $\mathbf{x_{ac}}$

Parameter	Variable	Value
Base Wing Aerodynamic Center	$x_{ac,w}$	0.31 MAC
Fuselage Nose to Wing MAC	l_{fn}	*Calculated from geometry*
Nacelle Position Factor	k_n	−2.5
Number of Nacelles	$n_{nacelles}$	2
Nacelle Maximum Diameter	d_{nac}	1.66 m
Nacelle to Wing MAC distance	l_n	*Calculated from geometry*

Table 10: Fundamental parameters used for the Cruise Aerodynamic Center shift (x_{ac}) calculation.

The total aircraft aerodynamic center for the cruise configuration, x_{ac}, is determined by taking the isolated wing's aerodynamic center and applying the destabilizing forward shifts caused by the fuselage and the engine nacelles.

The calculation relies on the aircraft-less-tail lift curve slope ($C_{L_{\alpha_{A-h}}}$), which was determined to be $6.5795\ \text{rad}^{-1}$ in a previous step, alongside wing and fuselage geometric parameters defined in Table 8.

First, the fuselage contribution is split into two parts to account for the physical dimensions and the wing sweep interference:

$$\Delta x_{ac,f} = -\frac{1.8 b_f h_f l_{fn}}{C_{L_{\alpha_{A-h}}} S_w \bar{c}_w} + \frac{0.273 b_f c_{gw}(b_w - b_f)\tan\Lambda_{w,c/4}}{(1+\lambda_w)\bar{c}_w^2(b_w + 2.15 b_f)} \tag{17}$$

Next, the forward shift caused by the engine nacelles is computed:

$$\Delta x_{ac,n} = n_{nacelles}\frac{k_n d_{nac}^2 l_n}{S_w \bar{c}_w C_{L_{\alpha_{A-h}}}} \tag{18}$$

The total normalized aerodynamic center location in cruise is the sum of these components:

$$x_{ac} = x_{ac,w} + \Delta x_{ac,f} + \Delta x_{ac,n} \tag{19}$$

Substituting the intermediate values and geometric parameters yields a final high-speed aerodynamic center of $x_{ac} \approx 0.1462$ MAC.

2.5.2 Controllability

Controllability refers to the ability of the aircraft to be trimmed, i.e., to maintain steady flight at a given condition using the available control authority. This is represented by the controllability (or trim) line, which defines the forward limit of the CG position for which the aircraft can be trimmed, given by:

$$\bar{x}_{cg} = \bar{x}_{ac} - \frac{C_{m_{ac}}}{C_{L_{A-h}}} + \frac{C_{L_h}}{C_{L_{A-h}}} \frac{S_h l_h}{S\bar{c}} \left(\frac{V_h}{V}\right)^2 \tag{20}$$

In this expression, $C_{m_{ac}}$ is the pitching moment coefficient about the aerodynamic center of the aircraft. The term C_{L_h} denotes the lift coefficient of the horizontal tail. The coefficient $C_{L_{A-h}}$ represents the lift curve slope of the complete aircraft minus the horizontal tail. All other variables are defined as previously described in Equation 10. In order to utilize Equation 20, it is necessary to determine the parameters x_{ac}, $C_{m_{ac}}$, $C_{L_{max}}$, and the horizontal tail downforce limit C_{L_h} for the most critical flight condition. For longitudinal controllability considerations, this condition is typically the landing approach state. During this phase, the aircraft operates at low speeds with high-lift devices fully deployed, which generates the maximum nose-down pitching moment that the horizontal stabilizer must successfully counteract.

1. $\mathbf{C_{L_{A-h}}}$

   For controllability considerations, the aircraft must be evaluated at its most critical flight condition. This occurs during the landing approach, where the aircraft operates at low speeds and the high-lift devices (flaps) are fully deployed. This configuration generates the maximum possible nose-down pitching moment that the horizontal stabilizer must successfully counteract.

   Because the aircraft is operating at this aerodynamic limit, the lift coefficient of the aircraft without the horizontal tail, $C_{L_{A-h}}$, is simply evaluated as the absolute maximum lift coefficient of the wing:

   $$C_{L_{A-h}} = C_{L_{max}}$$

   Based, this gives a constant maximum value of $C_{L_{A-h}} = 2.6$.

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2. $\mathbf{x_{ac}}$ (Landing)

   The aerodynamic center must be recalculated for the landing phase to account for the absence of high-speed compressibility effects. Unlike the cruise condition where high Mach numbers shift the aerodynamic center aft, the landing approach occurs at an effectively incompressible speed ($M \approx 0$).

   This removes the Prandtl-Glauert compressibility boost, setting $\beta = 1.0$. Consequently, the isolated wing's lift curve slope decreases to $C_{L_{\alpha_w}} = 4.4904$ rad^{-1}, and the theoretical base wing aerodynamic center shifts forward to $x_{ac,w} = 0.2600$ MAC.

   Because the wing generates proportionally less lift relative to the invariant physical geometry of the fuselage and engine nacelles, these forward components exert a stronger destabilizing influence. Re-evaluating the shift equations with the incompressible lift curve slopes yields a significantly more forward total aerodynamic center for the landing configuration:

   $$x_{ac,land} = x_{acw,land} + \Delta x_{ac,f,land} + \Delta x_{ac,n,land} \approx 0.0266 \text{ MAC} \tag{21}$$

3. $\mathbf{C_{m_{ac}}}$

Parameter	Variable	Value
Airfoil Zero-Lift Pitching Moment	C_{m_0}	-0.08
Airfoil Zero-Lift Coefficient	C_{L_0}	0.05
Wing Quarter-Chord Sweep	$\Lambda_{w,c/4}$	26.8°
Wing Area	S	77.4 m^2
Flap Start Position	y_{start}	1.289 m
Flap End Position	y_{end}	9.79 m
Flap Chord Extension Ratio	$\Delta c_f/c_f$	0.4
Double-Slotted Flap Lift Factor	–	1.6
Empirical Flap Moment Factor 1	μ_1	0.175
Empirical Flap Moment Factor 2	μ_2	0.55
Empirical Flap Moment Factor 3	μ_3	0.055

Table 11: Fundamental parameters and empirical constants used for the Total Pitching Moment ($C_{m_{ac}}$) calculation.

The total zero-lift pitching moment dictates the inherent rotational tendency of the aircraft. Because the critical condition is the landing approach, the severe nose-down moment generated by the deployed flaps must be superimposed onto the inherent wing and fuselage moments.

The wing's corrected base pitching moment evaluates to $C_{m_{ac,w}} = -0.1061$, given by:

$$C_{m_{ac,w}} = C_{m_0} \frac{A_w \cos^2 \Lambda_{w,c/4}}{A_w + 2\cos \Lambda_{w,c/4}} \tag{22}$$

The fuselage contribution depends on its length and height relative to the wing, evaluating to $\Delta C_{m_{ac,fus}} = -0.0244$:

$$\Delta C_{m_{ac,fus}} = -1.8 \left(1 - \frac{2.5 b_f}{l_f}\right) \frac{\pi b_f h_f l_f}{4 S \bar{c}} \frac{C_{L_0}}{C_{L_{\alpha_{A-h,land}}}} \tag{23}$$

To determine the pitching moment increment from the deployment of flaps, the proportion of the wing area affected by the flaps ($\frac{S_{wf}}{S}$) must first be evaluated. Assuming a linear taper, the local chord at any spanwise station y is:

$$c(y) = c_{rw} - \frac{2(c_{rw} - c_{tip})}{b_w} y \tag{24}$$

The flap area ratio is calculated by dividing the area of the flapped trapezoidal segment by the area of the half-wing:

$$\frac{S_{wf}}{S} = \frac{(c_{start} + c_{end})(y_{end} - y_{start})}{(c_{rw} + c_{tip})\frac{b_w}{2}} \tag{25}$$

where $c_{start} = c(y_{start})$ and $c_{end} = c(y_{end})$.

Next, the extended flap chord ratio, $\frac{c'}{\bar{c}}$, is evaluated based on the physical chord extension (Δc_f) and the mean aerodynamic chord ($\bar{c}$):

$$\frac{c'}{\bar{c}} = \frac{\Delta c_f + \bar{c}}{\bar{c}} \tag{26}$$

With the extended chord ratio determined, the maximum section lift coefficient increment ($\Delta c_{l_{max}}$) generated by the double-slotted flaps can be estimated. For double-slotted configurations, this increment is proportional to the extended chord ratio multiplied by an empirical factor of 1.6:

$$\Delta c_{l_{max}} = 1.6 \left(\frac{c'}{\bar{c}}\right) \tag{27}$$

This section lift increment, along with the extended chord ratio and the area proportion, is then utilized to calculate the quarter-chord pitching moment increment, which must subsequently be shifted to the landing configuration's aerodynamic center ($\bar{x}_{ac,land}$) using $C_{L_{max}}$. This results in a massive nose-down increment of $\Delta C_{m_{ac,HLD}} = -0.5382$:

$$\begin{aligned} \Delta C_{m,1/4} = \mu_2 &\left(-\mu_1 \Delta c_{l_{max}} \frac{c'}{\bar{c}} - \left(C_{L_{max}} + \Delta c_{l_{max}} \left(1 - \frac{S_{wf}}{S}\right)\right) \frac{1}{8} \frac{c'}{\bar{c}} \left(\frac{c'}{\bar{c}} - 1\right)\right) \\ &+ 0.7 \frac{A_w}{1 + 2/A_w} \mu_3 \Delta c_{l_{max}} \tan \Lambda_{w,c/4} \end{aligned} \tag{28}$$

$$\Delta C_{m_{ac,HLD}} = \Delta C_{m,1/4} - C_{L_{max}} (0.25 - \bar{x}_{ac,land}) \tag{29}$$

Summing these contributions yields the total pitching moment in the landing configuration:

$$C_{m_{ac}} = C_{m_{ac,w}} + \Delta C_{m_{ac,fus}} + \Delta C_{m_{ac,HLD}} \tag{30}$$

Substituting the values yields the final total pitching moment $C_{m_{ac}} \approx -0.6688$.

4. $\mathbf{C_{L_h}}$

   To trim the aircraft against the extreme $C_{m_{ac}}$ generated by the flaps, the horizontal stabilizer must exert its maximum possible downforce. The maximum horizontal tail downforce coefficient, C_{L_h}, represents the absolute aerodynamic limit of the tail before stalling.

   In this methodology, it is estimated empirically based on the tail's aspect ratio:

$$C_{L_h} = -0.35 A_h^{1/3} \tag{31}$$

   Substituting the updated tail aspect ratio ($A_h \approx 5.316$) yields the fixed downforce limit required to hold the nose up $C_{L_h} \approx -0.6115$.

Parameter	Sub-Component	RJ85
Cruise speed	–	$247.34\,\mathrm{m\,s^{-1}}$ (M 0.82)
Approach speed	–	$55.60\,\mathrm{m\,s^{-1}}$ (108.1 kn)
Tail-wing speed ratio (V_h/V)	–	0.95
Tail Lift Curve Slope ($C_{L_{\alpha_h}}$)	–	$4.9111\,\mathrm{rad^{-1}}$
Aircraft-less-tail Lift Curve Slope	**Total** ($C_{L_{\alpha_{A-h}}}$)	$6.5795\,\mathrm{rad^{-1}}$
	Wing ($C_{L_{\alpha_w}}$)	$6.1546\,\mathrm{rad^{-1}}$
	Fuselage Interference	$0.4249\,\mathrm{rad^{-1}}$
Wing downwash gradient ($\frac{d\epsilon}{d\alpha}$)	–	0.3232
Aerodynamic Center (Cruise)	**Total** (x_{ac})	0.1462 **MAC**
	Wing	0.3100 MAC
	Fuselage Shift	−0.2710 MAC
	Nacelles Shift	0.1072 MAC
Aerodynamic Center (Approach)	**Total** ($x_{ac,land}$)	0.0266 **MAC**
	Wing	0.2600 MAC
	Fuselage Shift	−0.3406 MAC
	Nacelles Shift	0.1072 MAC
Max Tail Lift Coefficient (C_{L_h})	–	−0.6115 (Fixed Stab.)
Zero Lift Pitching Moment (Approach)	**Total** ($C_{m_{ac}}$)	−0.6688
	Wing ($C_{m_{ac,w}}$)	−0.1061
	Fuselage ($\Delta C_{m_{ac,fus}}$)	−0.0244
	Flaps ($\Delta C_{m_{ac,HLD}}$)	−0.5382

Table 12: Summary of stability and controllability parameters for the RJ85 reference aircraft.

3 Analysis of CRJ-1000 with Modifications

3.1 General Infromation about CRJ-EXX

The CRJ-EXX is a modern evolution based on the already existing CRJ-1000 described in subsection 2.1, which intends to modernise the aircraft by implementing a new hybrid system. Design modifications have been done by Aerospark and are summarised in the following list:

1. Engine nacelle length and diameter have increased by +20, while keeping weight and center of gravity of propulsion system consistent with original aircraft
2. Innovative materials reduce wing weight by 9% and fuselage weight by 7%
3. Electric part of hybrid system is powered by two battery packs: one located in the forward part of the forward cargo hold (2025kg and volume of 1300l) and the second one located in the most aft part of the aft cargo hold (2475kg and 1600l)

4. Last three rows of passenger seats are removed
5. MTOW remains the same as original CRJ-1000, but fuel quantity is different as a result of modifications
6. New winglet system improves the effective Aspect Ratio by +25%
7. New High-Lift Devices implemented, raising the maximum coefficient of lift by +25%
8. Landing gear raised by $+30cm$ to guarantee safe battery recharging operations

As can be seen, the modifications are thorough, thus influencing a multitude of plance characterstics, including stability and maneuverability. These changes also influence the weight distribution of the aircraft. The updated table of aircraft parameters can be seen in Table 13:

Table 13: Parameters changed in CRJ-EXX configuration vs standard CRJ-1000

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
Payload & Cabin			**Fuselage & Cockpit Dimensions**		
Front Cargo CG Position	8.22	m	Fuselage CG Ratio	0.45	–
Aft Cargo CG Position	20.16	m	Reference z-Position	4.0	m
Number of Seat Rows	22	–	**Landing Gear Options** (*Wheels*)		
Row 1 Position	7.1	m	Nose Gear Station	2.126	m
Seat Pitch	0.79	m	Main Gear Station	22.10	m
Flight Characteristics & Limits			Main Gear Span (Half-Track)	3.0	m
Cruise Mach Number	0.82	–	Number of Main Gears	2	–
Cruise Altitude	9,000	m	Tail Strike Angle	15	deg

3.2 CG analysis and Loading Diagram for CRJ-1000 with Modifications

3.3 Stability and Control analysis for CRJ-1000 with Modifications

4 Team Organization and Reflection

5 Conclusion

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